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MILITARY SPECIFICATION

POWER PLANT, UTILITY: MUST HOSPITAL SYSTEM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a skid-mounted, self-contained, gas-turbine-engine-driven utility power plant which may be operated with a chemical-biological-radiological (CBR) collective protector. This utility power plant shall provide electrical power, air conditioning, heating, water, compressed air, and vacuum requirements for certain increments of the Army Surgeon General's Medical Unit, Self Contained, Transportable (MUST) hospital system. The specification also covers a winterization kit which, when installed, extends the low temperature operating range of the utility power plant from minus 25° F to minus 65° F.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS
FEDERAL

BB-F-1421	- Fluorocarbon Refrigerants.
QQ-S-781	- Strapping, Steel, and Seals.
VV-F-800	- Fuel Oil, Diesel.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-1055	- Barrier Material, Waterproofed Flexible.

FSC 6115

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PPP-P-40 - Packaging and Packing of Hand Tools.
PPP-T-60 - Tape: Packaging, Waterproof.
PPP-T-75 - Tape, Preservation and Sealing.

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MIL-P-116 - Preservation-Packaging, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant, (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514 - Plates, Identification, Instruction and Marking, Blank.
MIL-T-704 - Treatment and Painting of Materiel.
MIL-G-3056 - Gasoline, Automotive, Combat.
MIL-E-5607 - Engine, Gas Turbine, Preparation for Storage and Shipment of, Process for.
MIL-T-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-L-6081 - Lubricating Oil, Jet Engine.
MIL-C-45662 - Calibration System Requirements.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-705 - Generator Sets, Engine-Driven, Methods of Tests and Instructions.
MIL-STD-781 - Reliability Tests: Exponential Distribution.

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MIL-STD-810	- Environmental Test Methods.
MIL-STD-889	- Dissimilar Metals
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Army Materiel.

DRAWINGS
ME

TA13217E3800	- Utility Element, Self-Contained, Transportable, Type A.
TA13217E4010	- Winterization Kit, Utility Element, Self Contained, Transportable, Type A.
TA13217E4450	- Special Tools and Equipment.

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AIR MOVING AND CONDITIONING ASSOCIATION (AMCA)

Bulletin No. 210 - Standard Test Code for Air Moving Devices.

(Application for copies should be addressed to the Air Moving and Conditioning Association, 205 West Touhy Avenue, Park Ridge, IL 60068.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

Y32.2 - Graphic Symbols for Electrical and Electronics Diagrams.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

Standard 37 - Methods of Testing for Rating Unitary Air Conditioning Equipment.

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(Application for copies should be addressed to the American Society of Heating, Refrigerating and Air-Conditioning Engineers, 345 East 47th Street, New York, NY 10017.)

COUNTY OF LOS ANGELES - AIR POLLUTION CONTROL DISTRICT

Rule 66 - Organic Solvents.

(Application for copies should be addressed to the County of Los Angeles - Air Pollution Control District, 434 South San Pedro Street, Los Angeles, CA 90013).

3. REQUIREMENTS

3.1 Description. The utility power plant (hereinafter called "utility element") shall be in accordance with Top Assembly TA13217E3800, and as specified herein. The utility element shall be completely self-contained, requiring only fuel for the gas turbine engine and potable water for the water system. The necessary electrical cables, air conditioning ducts, water hoses, and auxiliary equipment required to provide utility services to inflatable or expandable shelters of a MUST hospital system shall be furnished and stored within the utility element. The maximum weight of the utility element fully equipped, including stowed equipment and a full charge of refrigerant and oil, shall not exceed 4700 pounds.

3.1.1 Winterization kit. When specified (see 6.2), a winterization kit shall be furnished with the utility elements. The winterization kit shall be in accordance with Top Assembly TA13217E4010 and as specified herein. The kit shall consist essentially of a heat exchanger, nickel-cadmium batteries, fuel heating provisions, and other necessary hardware and instructions for installation.

3.1.2 Special tools and equipment. When specified (see 6.2), special tools and equipment in accordance with Top Assembly TA13217E4450 shall be furnished.

3.1.3 Drawings. The drawings forming a part of this specification are end product drawings. No deviation form the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

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3.2 First article (preproduction model). The contractor shall furnish two or more utility elements as specified (see 6.2), including one winterization kit, when applicable, for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods will produce utility elements that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examination and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and test, as specified (see 6.2). The winterization kit shall be installed in the sequence indicated in Table II.

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more utility elements as specified (see 6.2) for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.4.1 Corrosion-resistant materials. Corrosion-resistant (stainless) steel, copper, anodized commercially pure aluminum alloys which do not contain more than 0.25 percent copper, and brass, bronze, beryllium-copper, copper-nickel, and nickel-copper alloys are considered corrosion-resistant materials for use on the utility element, as distinguished from corrosion-resistant treatments.

3.5 Environmental conditions.

3.5.1 High temperature. The utility element shall be capable of being stored at an ambient temperature of 160° F for 48 hours without damage or impaired performance after storage. The gas turbine of the utility element shall be capable of starting and operating for a short period of time (3 minutes minimum) at 155° F. The utility element shall be capable of starting and operating continuously in an ambient temperature of 140° F when loaded in accordance with 3.6.2.2 (a), except that the 400 Hz external output shall be reduced to 24 kW.

3.5.2 Low temperature. The utility element shall be capable of being stored at an ambient temperature of minus 65° F for a period of 48 hours

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without damage or impaired performance resulting from storage. A nonwinterized utility element shall be capable of starting and operating on the heating cycle without the application of external heat at an ambient temperature of minus 25° F. When the winterization kit is installed on a nonwinterized element, the utility element shall be capable of starting and operating on the heating cycle at an ambient temperature of minus 65° F.

3.5.3 Atmospheric pressure. The utility element shall start, operate and meet rated capacities at any atmospheric pressure from sea level to 10,000 feet, except that the cooling capacity will be reduced with altitude. The utility element shall also start and operate as specified herein after air transportation to 50,000 feet above sea level.

3.5.4 Rain. The utility element shall start and operate without malfunction when tested as specified in 4.6.3.18.3. Following the test, the utility element electrical system shall have an insulation resistance of not less than 50,000 ohms measured between current-carrying parts and current noncarrying parts.

3.5.5 Salt fog. The utility element shall operate without malfunction, and there shall be no corrosive damage or clogging or binding of moving parts when tested as specified in 4.6.3.18.5.

3.5.6 Humidity. The utility element shall start and operate without malfunction after testing as specified in 4.6.3.18.4.

3.6 Utility element outputs.

3.6.1 Environmental control system.

3.6.1.1 Cooling capacity. The utility element shall have a minimum cooling capacity of 240,000 Btu/hr at an ambient air temperature of 120° F dry bulb (db) and with an evaporator return air (recirculated air) of 90° F db, and 75° F wet bulb (wb) when operating against an external static pressure of 2 inches water gage (wg). Atmospheric pressure shall be sea level.

3.6.1.2 Recirculated airflow. When the utility element is operating at the conditions specified in 3.6.1.1, the recirculated airflow shall be not less than 5500 standard cubic feet per minute (scfm).

3.6.1.3 Heating capacity. The utility element, without the winterization kit installed, shall have a total heating capacity of not less

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than 3500,000 Btu/hr when operating against an external static pressure of 2 inches wg and in an ambient temperature of minus 25° F. With the winterization kit installed, the total heating capacity shall be increased to not less than 825,000 Btu/hr when operating against an external static pressure of 2 inches wg and in an ambient temperature of minus 65° F.

3.6.2 Electrical output.

3.6.2.1 Generator outputs (60 and 400 Hz). The output of the 60 Hz generator shall be not less than 12.5 kva (10 kw at 0.8 power factor (P. F.) and the output of the 400 Hz generator shall be not less than 120 kva (96 kw at 0.8 P. F.).

3.6.2.2 Utility element internal and external electrical loadings. The utility element shall be capable of continuously maintaining the following internal and external loads under all operating conditions specified in 3.7.

	Internal Loading		External Loading			
				Type of Power		
Loading	Utility Element System Operation	Kw at 0.8 P.F.	Element Receptacle	Volt	Phase	Frequency (Hz)
(a)	Air Conditioning 1 and 2; cold and hot water pumps operating	30	Power output	120/208	3	400
		10	Power output	120/240	1	60
(b)	Heating 1 and 2; cold and hot water pumps operating	30	Power output	120/208	3	400
		10	Power output	120/240	1	60
(c)	Ventilation fans 1 and 2; cold and hot water pumps operating	30	Power output	120/208	3	400
		30	Auxiliary	120/208	3	400
		30	Auxiliary	120/208	3	400
		10	Power output	120/240	1	60
(d)	One cooling system and cold and hot water pumps operating	30	Power output	120/208	3	400
		30	Auxiliary	120/208	3	400
		10	Power output	120/240	1	60
(e)	One heating system and cold and hot water pumps operating	30	Power output	120/208	3	400
		30	Auxiliary	120/208	3	400
		10	Power output	120/240	1	60
(f)	None	30	Power output	120/208	3	400
		30	Auxiliary	120/208	3	400
		30	Auxiliary	120/208	3	400
		10	Power output	120/240	1	60

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3.6.2.3 Motor loads. The 400 Hz electrical system shall be capable of starting the air conditioning system compressor while concurrently supplying a steady-state 3-phase load of 60 kw at 0.8 P.F. The 60 Hz electrical system shall be capable of starting a 2 horsepower, single-phase electric motor with an inrush of 58 amperes at 0.4 P.F. while concurrently supplying a steady-state single-phase load of 5 kw at 0.8 P.F.

3.6.3 Water supply.

3.6.3.1 Cold water. When operating at the conditions specified herein, the utility element shall provide cold water service at a rate of not less than 7.5 gallons per minute (gpm) and at a minimum pressure of 10 pounds per square inch gage (psig).

3.6.3.2 Hot water. When operating at the conditions specified herein, the utility element shall deliver a minimum of 6 gpm of hot water against a pressure of 10 psig. At any rate of hot water flow between zero to 1 gpm, the hot water temperature shall be maintained between 140° F and 160° F with continuous and intermittent usage.

3.6.4 Compressed air. The utility element shall provide 200 scfm of air at 1.5 psig at a temperature not exceeding 180° F when operating in an ambient temperature of 110° F to 120° F.

3.6.5 Vacuum. The utiliity element shall provide a suction source that will handle 10 scfm at 18 inches of mercury vacuum.

3.6.6 Engine fuel and oil consumption. The fuel consumption for the engine shall not exceed the value shown on Drawing D13217E4331. The oil consumption for the engine shall not exceed 0.04 pounds per hour under any operating conditions specified herein.

3.7 Starting and operation.

3.7.1 Starting and loading. The engine of the utility element shall start within 60 seconds after cranking is initiated without the use of external heat at any temperature from minus 25° F to plus 155° F. With the winterization kit installed, the utility element shall start within 60 seconds after cranking is initiated at any temperature from minus 65° F to plus 155° F. The "ready to load" light shall glow and the utility element shall be capable of being loaded within 90 seconds after initiation of cranking and maintaining any load specified in 3.6.2.2 in any ambient temperature from minus 25° F (minus 65° F winterized) to plus 120° F.

3.7.2 Fuel boost pump. The fuel boost pump shall operate when the turbine air intake door is open, with the fuel boost pump switch on and master switch in stop. The fuel boost pump shall also operate when the engine is running.

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3.7.3 Starting from external batteries. The utility element shall be capable of being started from an external power source supplied by a standard military vehicle 24-volt battery.

3.7.3.1 Reversed dc polarity. The engine shall not start from an external power source when the polarity of the dc power source is reversed nor shall the utility element be damaged from the reversed polarity. The polarity light shall glow when the battery polarity is reversed.

3.7.4 Emergency operation of engine. Emergency operation shall override all safety control devices except engine overspeed (see 3.7.9.1).

3.7.5 Standby operation. When two units are connected for standby operation in accordance with the operating instructions, the non-operating element (standby) shall start upon a failure of the operating element. The failed operating element shall give a 24 v dc signal to the standby connector and the standby element shall start when a 24 v dc signal is applied to the standby connector. The standby element shall also provide power output at both the 60 Hz and 400 Hz output connectors after starting.

3.7.6 Engine starting interlocks. The engine shall not start or operate if the turbine air intake door is closed. Also, the engine shall not start if the 400 Hz or 60 Hz load contactor switches are in the closed position.

3.7.7 Remote power. The utility element shall be capable of supplying power to one or two nonoperating units for the purpose of operating the recirculating fans of the nonoperating units.

3.7.8 Stopping. The utility element shall stop following the prescribed procedures of the operating instructions.

3.7.9 Safety and protection controls.

3.7.9.1 Overspeed. The engine shall shut down and coast to a stop when an overspeed condition is simulated in accordance with 4.6.3.3.1.

3.7.9.2 Low oil pressure. The engine shall shut down and coast to a stop when a low pressure condition is simulated in accordance with 4.6.3.3.2.

3.7.9.3 Turbine exhaust temperature. The turbine exhaust temperature shall not exceed 1260° F under any loading condition specified herein.

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3.7.9.4 Loss of fuel. The engine shall not crank without a fuel supply and shall shut down without damage when the fuel supply is depleted during operation. A fuel supply light shall indicate a lack of fuel.

3.7.9.5 Turbine combustion air restriction. When the turbine combustion air is restricted sufficiently to cause a vacuum of 12 inch water gage (w.g.) ± 1.5 inch w.g. in the turbine intake plenum, a warning shall be given of the impending operating problem by means of a light on the instrument panel. If the restriction continues to increase, the turbine shall stop when the turbine inlet pressure is 17.5 inch w.g.± 2 inch w.g.

3.8 Safety. All exposed parts that are subject to extreme temperature pressures, or that are energized electrically or are subject to explosion, and all rotating or reciprocating parts that are of such nature or so located as to become a hazard to operating personnel shall be insulated, enclosed or fully guarded. Safety features shall include necessary indicating and warning devices, interrupters, circuit breakers, relays temperature and pressures switches, relief valves, and protective enclosures and guards. A terminal stud sized to accommodate a No. 6 AWG solid wire shall be provided on the enclosures in the vicinity of the main output or receptacle panel for the purpose of grounding the utility element.

3.9 Sound level requirements. The sound level of the utility element when tested in accordance with 4.6.3.20 shall not exceed the following steady-state noise limits in any position of measurement:

Octave Band Limits (Hz)	Center Frequency (Hz)	Noise Limits (dB)
44-88	63	90
88-177	125	87
177-354	250	88
354-707	500	80
707-1414	1000	77
1414-2828	2000	74
2828-5657	4000	73
5657-11314	8000	73

Hearing protection need not be supplied with the utility element. If the utility element can be operated at a level that exceeds 85 dB(A), the appropriate discussion of the noise hazard in MIL-STD-1474 (MANUALS Paragraph) shall include the requirement for hearing protection, the noise level of the item, the distance from the item at which 85 dB(A) will always be met, and what operational or system configuration situations

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will reduce the level to 85 dB(A). Hazard signs, if required, shall be in accordance with and shall be located where required by the drawings.

3.10 Mobility and usage.

3.10.1 Roadability. The utility element when tested as specified in 4.6.3.18.6 shall withstand the vibration and sudden acceleration incident to transportation without evidence of structural damage, visual misalinement of components, dislocation of components, or breakage of lines, and shall start and operate as specified herein.

3.10.2 Transportability. The utility element shall sustain no damage and shall be operable immediately after being transported by the following methods:

(a) Military van or truck and demountable running gear on rough cross-country roads at variable speeds up to 20 mph.
(b) Military or commercial rail system.
(c) Air transport by helicopter or aircraft.
(d) Ocean or inland water transportation methods.
(e) Forklift truck.

3.10.3 Transit drop. The utility element shall withstand and absorb shock without structural damage, visual misalinement of components, malfunctioning of components or controls, or any other irregular operation when tested as specified in 4.6.3.18.7.

3.10.4 Inclination. The utility element shall start and operate in each of four positions, front to rear and side to side, without malfunction at a 15-degree positive or negative displacement of the front and rear axis and with a 10-degree displacement to either side. The utility element, while nonoperational, shall be capable of being tilted 45 degrees in any direction without damage, spillages or other adverse effects that would make the utility element inoperable.

3.10.5 Lifting and towing attachments. Lifting rings shall be provided at each of the four upper corners of the enclosure and shall withstand without damage or permanent deformation a load of at least 4-1/2 times its proportionate share of the weight of the fully equipped utility element. Four towing eyes, two located on the front and rear ends of the bottom of the utility element, shall be provided. The lifting and towing attachments shall also serve as tiedown fittings. Each attachment shall withstand without damage to the utility element

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its proportion of the following extreme shock-loading conditions during transport: 8 G forward, 8 G rearward, 2-1/4 G upward, and 1-1/2 G sideward, where G equals the weight of the fully equipped utility element.

3.11 Electrical system

3.11.1 Steady-state ac voltage regulation. The ac voltage shall be regulated within the range of plus or minus 1 percent of the voltage as set. This regulation shall be maintained when balanced loads are applied from no load to maximum continuous rated load under all ambient and operating conditions specified herein without changing the voltage setting. Continuous rated load shall be as specified in 3.6.2.2(c). Regulation shall be plus or minus 1 percent for the 400 Hz system and plus or minus 2.5 percent for the 60 Hz system as the load is changed from no load to full load and vice versa, once the generating system has reached stabilized operating temperature conditions.

3.11.2 Wave form. With the alternator operating at rated voltage and frequency under control of its exciter and voltage regulator, and at any balanced unity power factor at 0.8 P.F. (lagging) load between no load and rated kilowatts, the following requirements shall apply to line-to-line and line-to-neutral voltage wave forms: (a) the deviation factor shall not exceed 5 percent, (b) no single harmonic shall exceed 3 percent of the fundamental, and (c) rms total harmonic content shall not exceed 5 percent. This requirement is based on the use of linear (nonsaturating) reactors to obtain the lagging power factor load component. The crest factor shall be 1.41 plus or minus 0.1.

3.11.3 Transient voltage. The voltage shall not dip below 65 percent of rated voltage when subjected to the motor starting load specified in 3.6.2.3 and shall recover to within plus or minus 5 percent of steady state limits within 1.0 second for a 400 Hz generator and within 2.5 seconds for a 60 Hz generator. Application or rejection of rated load in one step shall not cause the voltage to change by more than 25 percent of its initial value. The voltage shall reach its final value within 1.0 second for the 400 Hz generator and within 2.5 seconds for the 60 Hz generator and shall stabilize at this value with not more than one perceptible overshoot.

3.11.4 Voltage adjustment. The voltage level shall be adjustable between 115 and 125 volts by means of an accessible adjustable rheostat contained within the regulator. The adjustment shall be such that, once set, operation of the generator set will not cause the voltage level to vary outside of the limits specified in 3.11.1.

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3.11.5 Frequency limits. When the governor is set at the nominal speed set point, the generator frequency shall remain within plus or minus 2.5 percent for all loads from zero to full load (including transient load changes) throughout the ambient operating temperature and pressure range. When the governor is set for local altitude, the nominal performance shall be plus or minus 0.5 percent for any set load from no load to full load.

3.11.6 Frequency transient. The frequency shall not change sufficiently to cause the underfrequency protection to disconnect the generator from the load bus and shall recover to the set point within 1 second with the application or removal of the load specified in 3.6.2.2(c).

3.11.7 Electrical system protection.

3.11.7.1 Generator overload. The generator overload protector shall disconnect the load from the generator when the generator is overloaded or a short circuit develops. The overload response time versus the percent of rated load as specified in 3.6.2.1 shall be as follows:

60 Hz Generator

Percent rated load	Response
0-110 Percent	Shall not trip
150 Percent	Shall trip in 30 to 120 seconds
Short circuit	Shall trip in 1.0 to 2 seconds

400 Hz Generator

Percent rated load	Response
0-110 percent	Shall not trip
150 percent	Shall trip in 15 to 25 seconds
Short circuit	Shall trip in 1.0 to 2 seconds

3.11.7.2 Underfrequency protector 400 Hz generator. The underfrequency protector shall disconnect the generator from the load bus when the frequency drops to 374 plus or minus 4 Hz for 1.5 to 2.0 seconds.

3.11.7.3 Undervoltage. The individual 400 Hz generator or 60 Hz generator shall be disconnected from the load when the line to neutral voltage drops to 90 plus or minus 2 volts or less and remains there for at least 4.5 seconds.

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3.11.7.4 Overvoltage.

3.11.7.4.1 Generator, 60 Hz. The generator shall not be disconnected from the load with a single-phase step input voltage of 129 volts line-to neutral and shall be disconnected within 0.5 to 10 seconds with a single-phase step input voltage of 135 volts line-to-neutral.

3.11.7.4.2 Generator, 400 Hz. The generator shall not be disconnected from the load with a 3-phase step input voltage of 129 volts line-to-neutral and shall be disconnected within 0.5 to 10 seconds with a 3-phase step input voltage of 135 volts line-to-neutral.

3.11.7.5 Anticycling. The electrical control system shall provide a means to prevent cycling of any ac portion when a fault exists and the load switch is held in the reset position. The control assembly or protective device shall reset once after a fault and, if the fault remains, the protective panel shall trip and remain tripped as long as the load switch shall be held in the reset position.

3.11.8 Electrical instruments. All electrical indicating instruments shall have an accuracy equal to or better than plus or minus 2 percent of the full scale value.

3.11.9 Battery charger. The battery charger shall charge a battery having any condition of discharge without injury to the battery and without evaporating an appreciable amount of water. A fully charged battery shall be maintained at 28.0 plus or minus 1.0 volts dc.

3.11.10 Lights. The utility element and control panel shall incorporate lights that will glow and indicate equipment operation, equipment failure and malfunction.

3.11.11 Bleed air. The turbine shall be capable of supplying bleed air as required for the heating system, de-icer manifold and ejector to produce compressed air for inflating shelters and a vacuum. The maximum acceptable contamination level of the bleed air shall in no case exceed 2 parts per million of nonvolatile residue.

3.11.12 Phase sequence. The phase sequence of all 3-phase power receptacles and connectors for any utility element rotating equipment shall be clockwise.

3.12 Environmental control system. The utility element shall be capable of ventilating (recirculating air without heating or cooling), cooling (air conditioning) the recirculated air, or heating the recirculated air. The environmental control system shall consist of two independent

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ventilating systems, two independent heating systems and two independent air conditioning systems. The same recirculating fans and air recirculating system shall be used for all modes of operation.

3.12.1 Environmental control system equipment operation. Selection of the ventilating or heating or air conditioning function shall be made with the mode select switch and the system selection shall be made with the heat system select, vent system select, or air conditioning system select switches. Equipment operation shall be in accordance with Table I. The direction of rotation of all fans shall be as indicated on each fan.

Table I. Environmental Control System Operation

System Select			Equipment operation							
			System No. 1 2/				System No. 2 2/			
Mode Select	Heat	Vent	Air Cond	Recir Fans	Cond Fans	Compr	Heat Valve 3/	Recir Fans	Cond Fans	Compr	Heat Valve
	1/										
Off	Any	Any	Any	Off	Off	Off	C	Off	Off	Off	C
Vent	Any	1	Any	On	Off	Off	C	Off	Off	Off	C
Vent	Any	2	Any	Off	Off	Off	C	On	Off	Off	C
Vent	Any	1&2	Any	On	Off	Off	C	On	Off	Off	C
Heat	1	1&2	Any	On	Off	Off	O3/	On	Off	Off	C
Heat	2	1&2	Any	On	Off	Off	C	On	Off	Off	O
Heat	1&2	Any	Any	On	Off	Off	O	On	Off	Off	O
Cool	Any	2	1	On	On	On	C	Off	Off	Off	C
Cool	Any	1	2	Off	Off	Off	C	On	On	On	C
Cool	Any	Any	1&2	On	On	On	C	On	On	On	C

1/ Any - Indicates the system select switch may be placed in any position.

2/ Equipment:
(a) Recir Fans - Recirculating fans (2 for each system).
(b) Cond Fans - Condenser fans (2 for each system).
(c) Compr - Compressor (1 per system).
(d) Heat Valve - Heat flow control valve (1 per system).

3/ C - Heat flow control valve closed.
O - Heat flow control valve open. Set system temperature control for maximum heat with inlet temperature below 90° F.

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3.12.1.1 Environmental control system and auxiliary power output. The environmental control system and auxiliary power output shall be interlocked as follows:

(a) Selection of No. 1 auxiliary power output receptacle will permit only No. 2 air conditioning system or heating system to operate.
(b) Selection of No. 2 auxiliary power output receptacle will permit only No. 1 air conditioning system or heating system to operate.
(c) Selection of both auxiliary power output receptacles will not permit any air conditioning or heating control system to operate.

3.12.2 Temperature control.

3.12.2.1 Heating. Each heating system shall have an independent control system. The heat range selector switch can be set in any of four positions: off, low, medium, and full. The maximum output in each position is limited to 0 percent, 25 percent, 50 percent, and 100 percent, respectively, of the rated heating capacity. Automatic temperature control shall be provided in any heat range selector setting by opening or closing the bleed airflow control valve at the command of the return air temperature sensors for the heating system. The temperature control shall have a manually adjustable range from 0° F to 90° F.

3.12.2.2 Air conditioning. Each air conditioning system shall have an independent control system. The compressor, recirculating fans and condenser fans of each system shall operate continuously once started. The compressor bypass valve, refrigerant liquid line valve and the refrigerant de-icing valve shall cycle at the command of the air conditioning return air sensors. The temperature control system shall have a range from 60° F to 90° F and a temperature differential not exceeding plus or minus 3° F.

3.12.2.3 Air conditioning system. The vapor compression cycle shall be used for cooling. The refrigerant employed shall conform to BB-F-1421, Type 114. The amount of refrigerant charge for System No. 1 (lower) shall be 39 pounds of refrigerant R-114 and for System No. 2 (upper) shall be 36 pounds of refrigerant R-114. The refrigeration oil shall be the type recommended by the compressor manufacturer; and the amount shall be 590 cubic centimeters (cc) for System No. 1 and 560 cc for System No. 2. The refrigerant moisture content in the system shall not exceed 15 parts

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per million (ppm) by weight in the liquid phase. The evaporator coil shall not "ice up" or collect frost on the surface to the point where icing causes the airflow to be cut off. The entire refrigerant system shall be capable of withstanding internal pressure up to 150 percent of the refrigerant saturation pressure at 150° F without deformation of any components or without any detectable leak. All condensate from the air conditioner evaporator coil and any other water entering the environmental control system shall drain from the utility element. There shall be no moisture carryover leaving the conditioned air outlet with 2 inches of water external static pressure.

3.12.3.1 Air conditioning safety controls.

3.12.3.1.1 Condenser air door interlock. If either the condenser air intake cover or the discharge cover is closed, the air conditioner system shall not operate.

3.12.3.1.2 Refrigerant high pressure cut-out switch. The high pressure cut-out switch for both air conditioning refrigerating systems shall be set to stop the compresor when the condensing pressure reaches 135 plus or minus 7 psig.

3.12.3.1.3 Compressor overload. The compressor motor shall be protected against damage resulting from locked rotor conditions and a running load which exceeds 100 amperes.

3.12.3.1.4 Fan motor overload. All fan motors shall be protected against damage resulting from locked rotor conditions.

3.12.4 Heating system.

3.12.4.1 Heating system safety controls.

3.12.4.1.1 High temperature air limit. The temperature of the recirculating air shall not exceed 225° F under any condition of operation specified herein.

3.12.4.1.2 Electric power failure. The turbine bleed air shall not flow into the heating system in the event of a 400 Hz electric power failure.

3.12.5 Fresh air. The utility element shall be equipped with four fresh air inlet doors as depicted on the appropriate drawings. Each door shall be manually adjustable so that the amount of fresh air admitted to the recirculating air system may be varied in any amount from zero to the maximum amount. All fresh air shall pass through the utility element air filter before leaving the utility element.

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3.13 Water system. When supplied with water at atmospheric pressure from an external source, an operating nonwinterized utility element shall be capable of supplying both hot and cold water and storing hot water at any ambient temperature from minus 25° F to plus 140° F. The range of performance for an operating winterized utility element shall be from minus 65° F to plus 140° F.

3.13.1 Low temperature operation. Freezing of water shall not occur in any part of the water system of the utility element (including supply hose and dual hot and cold water hose) at any of the conditions specified in 3.13.

3.13.2 Priming. Once the water system is in operation, the system shall maintain the prime.

3.13.3 Drainage. All components shall drain sufficiently to prevent freezing when the utility element is stored at low temperatures. All drains and outlets shall be accessible and without water traps. Turbine bleed air may be utilized to dry the water system.

3.14 Reliability and maintainability.

3.14.1 Reliability. The specified Mean-Time-Between-Failure (MTBF) shall be 840 hours when the utility element is tested as specified in 4.6.3.24.

3.14.2 Maintainability. The ratio of man-hours of scheduled and unscheduled maintenance performed during the entire test program to the hours of operation accumulated therein shall not exceed 0.08. The achieved availability shall not be less than 0.95.

3.15 Electromagnetic interference. The electromagnetic interference emission characteristics of the utility element shall conform to Table A-1 of MIL-STD-461 with Notice 1 when tested in accordance with 4.6.3.21 (see 6.8).

3.16 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, Composition I or II, as applicable (see 3.16.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

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When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.16.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of Composition I. When Composition II is used, the contractor shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 66, Air Pollution Control District, County of Los Angeles, CA.

3.17 Identification marking. The utility element shall be identified in accordance with MIL-STD-130.

3.18 Shipping data plate. The shipping data plate shall conform to MIL-P-514, Type III, Composition C, Grade A, Class 1 material. The plate shall depict the silhouette of the utility element in position suspended by a sling and indicating the center of gravity and the location and capacity of the lifting and towing attachments. The plate shall be attached by screws, bolts or rivets in a location where the plate will be both visible and legible.

3.19 Schematic diagrams. Schematic diagrams depicting the electrical circuitry refrigeration system, compressed air system, and hot and cold water systems shall be mounted on the inside surfaces of the door panels of the utility element. Electrical symbols shall be in accordance with ANSI Y32.2.

3.20 Instruction plates. Each utility element shall be equipped with instruction plates, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the utility element.

3.21 Treatment and painting. All parts shall be cleaned, treated and painted as indicated on the drawings, except as follows:

(a) Internal components previously painted by the contractor need not be repainted if they conform to the salt-fog requirements.
(b) Coils, refrigerant tubing, electrical components, acoustical insulation, or other components whose operation will be impaired by painting shall not be painted.
(c) Dissimilar metals shall be insulated from one another in accordance with MIL-STD-889.

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When not indicated on the drawings or included in (a), (b), or (c), parts shall be treated, cleaned, and painted in accordance with MIL-T-704.

3.22 Human factors. The characteristics of the utility element shall provide for operation by personnel in all type clothing and shall be designed in compliance with Section 4 of MIL-STD-1472.

3.23 Government-furnished property. The following property in the quantity indicated will be furnished by the Government (see 6.10):

Item No.	Description	Identification	Qty for ea. Power Plant, Utility
1.	Filter/separator, Liquid Fuel; Skid mounted, 15 GPM Capacity	NSN 4330-00-438-1460	1
2.	Extinguisher, Fire	NSN 4210-00-555-8837	1

3.24 Workmanship. All parts, components, and assemblies of the utility element including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. External surfaces shall be free from burrs, sharp edges and corners except when sharp edges and corners are required.

3.24.1 Metal fabrication. Sheet metal and structural members used in fabrication shall be straight and free from laminations. Sheet metal and structural members may be cut by shearing or sawing. All burrs and rough edges shall be removed. Corners shall be square and true. All bends shall be made to a radius which will not cause cracking, rippling or flattening, and all bends except for minor details shall be made by use of metal dies or fixtures to insure uniformity in size and shape.

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3.24.2 Bolted connections. Boltholes shall be punched or drilled and shall have the burrs removed. Washers and lockwashers shall be provided where necessary. All bolts, nuts, and screws shall be correctly torqued.

3.24.3 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the members.

3.24.4 Jigs and fixtures. Shop-fabricated components shall be assembled in steel jigs or frames and joined while held in position. Jigs and frames shall be designed to minimize distortion.

3.24.5 Interchangeability. All like components shall be interchangeable with respect to installation, performance and quality.

3.24.6 Refrigeration system. Extreme care shall be exercised during all phases of fabrication and assembly to eliminate contaminants from the refrigeration system such as residual solvents, moisture, metallic particles, flux, dirt, scale, and rust. This care shall include but shall not be limited to:

(1) Thoroughly cleaning all tubing with Dupont Freon TF Solvent Cleaning Agent or equivalent following fabrication;
(2) The thorough cleaning and degreasing of all regrigeration components;
(3) Sealing all tubing and other refrigeration components with shipping caps to prevent the entrance of contaminants after cleaning and the removing of their caps only at installation; and
(4) Maintaining a sealed refrigeration system during fabrication, whenever possible.

3.24.7 Water leakage. The condenser compartment floor and the refrigerant compartment floor shall not leak when tested in accordance with 4.6.3.17.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

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The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications, standards and drawings, as applicable.

4.1.2 Disassembly inspection. Failure of any examination or test of the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be cause for rejecting the preproduction model. Government-furnished property shall be excluded from the requirements of this paragraph.

4.1.3 End Item Final Inspection Requirement (EIFIR). Each utility element shall be inspected in accordance with the EIFIR specified on the drawings. One copy of the completed EIFIR shall be shipped with each utility element (see 6.4.1).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3)
(b) Initial production inspection (see 4.4)
(c) Quality conformance inspection (see 4.5)
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. Each utility element shall be examined as specified in 4.6.2. Presence of one or more defects shall be cause for rejection and for performing the inspection specified in 4.1.2.

4.3.2 Tests. Two or more utility elements shall be used to perform the tests specified in Table II. The tests shall be conducted either in a sequence as shown in a Government-approved test plan or as follows:

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(a) When two preproduction models are furnished, one utility element shall be subjected in sequence to those tests indicated in Column 2, "Prepro No. 1" and the other utility element shall be subjected in sequence to those tests indicated in Column 3, "Prepro No. 2."

(b) When more than two preproduction models are furnished, two utility elements shall be tested as specified in 4.3.2.a; the remaining utility elements shall be subjected only to Inspection Sequence 33 (Reliability). Failure of any test shall be cause for rejection and for performing the disassembly inspection specified in 4.1.2.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production utility elements will be selected at random by the Government from utility elements being produced from production tooling and will be subjected to the examination specified in 4.6.2 and the tests marked "X" in Column 5 of Table II to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production utility element shall not exclude the remaining utility elements from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production utility element to meet any requirement specified herein during and as a result of the inspection specified in 4.4 shall be cause for rejection of the initial production utility element(s) and shall be cause for refusal by the Government to continue acceptance of production utility elements until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on utility elements previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all utility elements accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

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4.5 Quality conformance inspection.

4.5.1 Examination. Each utility element shall be examined as specified in 4.6.2. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each utility element shall be tested as specified in Column 4 of Table II. Failure of any test shall be cause for rejection.

4.6 Inspection procedure.

4.6.1 Inspection schedule. The inspection schedule shall be as specified in Table II.

Table II. Inspection Schedule

Inspection sequence	Type of inspection: Prepro. No. 1	Propro. No. 2	Quality conformance	Initial prod and inspection comparison	Test	Requirement para.	Inspection para.
1	2	3	4	5	6	7	8
1	X	X	X	X	Examination	3.1, 3.4, 3.16, 3.17, 3.18, 3.19, 3.20, 3.21, 3.24.	4.6.2
2	X	X	X	X	Operation	3.7.1 to 3.7.3, 3.7.6 to 3.7.8, 3.7.9.3, and 3.12.1	4.6.3.2
3	X	X	X	X	Engine protection	3.7.9.1, 3.7.9.2, 3.7.4, 3.7.9.4, 3.7.9.3, 3.7.9.5	4.6.3.3

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Table II. Inspection Schedule cont'd)

1	2	3	4	5	6	7	8
4	X	X	X	X	Electrical protection	3.11.7	4.6.3.4
5	X	X	X	X	Electrical system check	3.6.2.2, 3.11.12, 3.11.10	4.6.3.5
6	X	X	X	X	Electrical system performance	3.11.1, 3.11.3, 3.11.4, 3.11.5, 3.11.9	4.6.3.6
7	X 1/	-	-	X	Generators	3.11.2	4.6.3.7
8	X	X	X	X	Instruments	3.11.8	4.6.3.8
9	X	X	X	X	Refrigeration system quality	3.12.3	4.6.3.9
10	X	X	X	X	Air conditioning system protection and controls	3.12.2.2, 3.12.3	4.6.3.10
11	X	X	X	X	Recirculating airflow check	3.12.3	4.6.3.11.1
12	X	X	X	X	Cooling system check	3.12.3, 3.6.1	4.6.3.11.2
13	X	-	-	-	Cooling capacity	3.12.3, 3.6.1	4.6.3.11.3
14	X	-	-	-	Evaporator coil frost	3.12.3	4.6.3.11.4
15	X	X	X	X	Heating overtemperature switch	3.12.4.1	4.6.3.12.1
16	X	X	X	X	Heating temperature control	3.12.2.1	4.6.3.12.2
17	X	X	X	X	Electric power failure during heat cycle	3.12.4.1	4.6.3.12.3
18	-	X	-	-	Heating capacity	3.6.1.3	4.6.3.12.4
19	X	X	X	X	Cold water flow rate	3.6.3.1, 3.13	4.6.3.13.1
20	X	X	X	X	Hot water flow rate	3.6.3.2, 3.13	4.6.3.13.2
21	X	X	X	X	Hot water temperature control	3.6.3.2	4.6.3.13.3
22	X	X	X	X	Hot water pump pressure switch	-	4.6.3.13.4
23	X 1/	-	-	X	Hot water capacity	3.6.3.2, 3.13.2	4.6.3.13.5
24	X	X	X	X	Compressed air	3.6.4	4.6.3.14

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Table II. Inspection Schedule (cont'd)

1	2	3	4	5	6	7	8
25	X	X	X	X	Vacuum system	3.6.5	4.6.3.15
26	-	X	-	-	Winterization kit	3.1.1	4.6.3.16
27	X	X	X	X	Water leakage	3.24.7	4.6.3.17
28	-	X 2/	-	X	Environmental	3.5.1, 3.5.2, 3.5.4, 3.5.5, 3.5.6, 3.10.2, 3.10.3, 3.12, 3.13	4.6.3.18
29	-	X 1/	-	-	Handling and positioning	3.10.4, 3.10.5	4.6.3.19
30	X	-	-	X	Sound level measurement	3.9	4.6.3.20
31	X	-	-	-	Electromagnetic interference	3.15	4.6.3.21
32	X	X	X	-	Production performance	3.6, 3.7 3.11, 3.12	4.6.3.22
33	X	X	-	-	Reliability	3.14.1	4.6.3.24
34	-	-	-	X	Maintainability	3.14.2	4.6.3.25
35	-	-	-	X	Human factors	3.22	4.6.3.26
36	-	-	-	X	Safety	3.8	4.6.3.27

NOTES

1/ Test may be conducted on either preproduction utility element.

2/ Winterization kit shall be removed at completion of environmental tests or when otherwise specified (see 6.2).

4.6.2 Examination. The utility element shall be examined as specified herein for the following defects:

101. Nonconformance to drawings.
102. Weight of the utility element not as specified.
103. Stowed accessory items missing.
104. External dimensions of the utility element not as shown on the drawings.
105. Fire extinguisher missing.
106. Hinged doors on the outside of the utility element that do not open and close or do not fasten as intended.

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107. Fasteners on the utility element outside panels that do not function as intended.
108. Covers missing from the conditioned air inlet and outlet connection.
109. Stowed equipment not stowed in accordance with instructions in storage compartments.
110. Hoisting rings (4), forklift points, wheeled transporter points, or skidding and tiedown rings not as shown on the drawings.
111. Panels or connections not as specified.
112. Door interlock switches not on all condenser doors and turbine air inlet door.
113. Remote temperature sensor connector missing.
114. Folding steps missing.
115. Schematic diagrams and instruction plates missing or not as specified.
116. Shipping data plate missing or not as specified.
117. Treatment and painting defects such as blisters, thin spots, lumps, scratches, runs, or sags.
118. Repair parts and maintenance tools missing (including special tools if applicable).
119. Instructional material missing or not as specified.
120. Any evidence of fuel, oil, refrigerant, water, or compressed air leaks.
121. Wires, cables or tubing that are pinched, twisted, exposed to fuel, oil, or parts at high temperatures, or that are contacting sharp corners, edges or moving parts.
122. Electrical wires or cables too short or too long to be neatly bundled together and supported.
123. Electrical wires or cables not securely supported with insulating clamps.
124. Wires not marked as specified.
125. Electrical soldered joints not made with a resin soldering flux or the flux not removed after soldering.
126. Electrical circuitry not protected from the effects of moisture or fungus growth.
127. Solder joints which have evidence of charring, burning, or other heat damage.
128. Soldered joints in which the solder completely obscures the configuration of the joint.
129. Cold solder joints which have a dull frosted "pile up" appearance.
130. Loose parts resulting from heating.
131. Welds which are cracked, porous or undercut, or which contain voids or gaps.

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132. Rivets not tight or not having the rivet head straight or tightly seated against their bearing surface.
133. Screw fasteners that are not tight so that there is relative movement possible between the attached parts.
134. Threaded parts or devices that show evidence of cross threading, detrimental or hazardous burrs or mutilation.
135. Washers or lockwashers missing.
136. Components and subassemblies not identified in accordance with MIL-STD-130.
137. Warping or bowing of structural members, mounts, or panels due to welding.
138. Evaporator and condenser coils not coated on the exterior heat transfer surfaces with a protective coating.
139. Provision not provided to drain all parts of the water system.
140. Electromagnetic interference suppression components or networks not identical to the approved preproduction model.
141. Fresh air inlet dampers cannot be set in all positions from closed to fully open.
142. Parts or materials which are fungus inherent not treated for fungus.
143. Pipe or tubes in contact with sharp corners or in contact with other pipes or tubes.
144. Electric wiring in contact with sharp corners.
145. Tubing bends which are flattened or rippled.

4.6.3 Tests.

4.6.3.1 Conditions. Unless otherwise specified herein, tests shall be conducted employing fuel conforming to MIL-T-5624, Grade JP-4 for the utility element. The utility element engine speed shall be set so that the electrical output frequency from the 400 Hz generator is 400 Hz plus or minus 0.5 percent under all steady load conditions. No braces, sealing materials, tiedown devices, or other items that are not inherently a part of the operating utility element shall be employed. Unless otherwise specified herein, no alinement of components, addition of refrigerant or other servicing shall be performed on the preproduction models or production utility elements after testing has been started. Components subject to corrosion shall not be treated with any treatment such as grease or oil in order to meet test requirements. All testing shall be conducted with instrumentation and apparatus specified in American Society of Heating Refrigerating and Air Conditioning Engineers (ASHRAE) Standard 37, MIL-STD-705, and as specified herein. Unless otherwise specified herein, all tests shall be conducted with all panels installed. When the utility element is located in a chamber for testing, the turbine combustion air and exhaust gases should not be restricted so as to impair turbine

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operation. The divider shall be placed in the recirculating air discharge chamber for all tests.

4.6.3.1.1 Calibration. All instruments shall be calibrated prior to testing and shall comply with the requirements or instructions of the applicable specifications. A system for the calibration of all measuring and test equipment shall be maintained in accordance with MIL-C-45662.

4.6.3.2. Operation. Check the utility element by following the operating instructions mounted inside the utility element instrument panel cover. All checks shall be conducted in an ambient temperature of 60° F to 90° F unless otherwise specified herein.

	Checks	Failure Criteria: Nonconformance to
(a)	Start with unit batteries, measure time to start and observe operation of fuel boost pump.	3.7.1, 3.7.2
(b)	Stop.	3.7.8
(c)	Start with external dc source	3.7.3
(d)	Start with external dc source and reversed polarity.	3.7.3.1
(e)	Stop unit; emergency shutdown.	3.7.8
(f)	Start with 400 Hz main contactor switch closed.	3.7.6
(g)	Start with 60 Hz load contactor switch closed.	3.7.6
(h)	Start the unit with the turbine air intake door closed (Note: If unit attempts to start, stop immediately).	3.7.6
(i)	Operate environmental system in ventilation, refrigeration and heating as required by Table I (Note: When checking the heat system(s), the air temperature entering the inlets may require cooling). Observe the operation of fans, compressors and valves.	3.12.1
(j)	Remote power operation, running unit.	3.7.7
(k)	Remote power operation, remote non-operating unit. In lieu of a running unit, an external source of 208 volt 3-phase 400 Hz power may be used.	3.7.7
(l)	Operate the utility element in the "cool" mode with the air conditioning system(s) and the auxiliary output switches set as follows:	

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Checks	Failure Criteria: Nonconformance to	
Air Conditioner Sys Select	Auxiliary 1	Switch 2
1	OFF	ON
2	ON	OFF
1	ON	OFF
2	OFF	ON
1 & 2	ON	ON

3.12.1.1

(m) Operate the utility element in the "heat" mode with the heating system(s) and the auxiliary output switches set as follows:

Heating Sys select	Auxiliary 1	Switch 2
1	OFF	ON
2	ON	OFF
1	ON	OFF
2	OFF	ON
1 & 2	ON	ON

3.12.1.1

(n) Standby operation. Stop the running unit by disconnecting the low oil pressure switch. After start of the standby unit, measure the voltage at the 60 Hz and 400 Hz output connectors. In lieu of using two units, the output signal of the running unit at the standby connector may be measured and an external 24 v dc signal may be applied to the standby unit. 3.7.5

(o) Operate the utility element with the 12-foot bleed air hose connected to the de-icer manifold. Determine if bleed air is blowing out all manifold holes. The lack of bleed air being ejected from any hole shall constitute failure of this test.

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4.6.3.3 Engine protection.

4.6.3.3.1 Engine overspeed. With the utility element operating, remove the electrical connector from the engine overspeed switch. Failure of the engine to stop shall constitute failure of this test.

4.6.3.3.2 Low oil pressure. With the utility element operating, remove the electrical connector from the engine low oil pressure switch. Failure of the engine to stop shall constitute failure of this test.

4.6.3.3.3 Emergency operation. With the utility element operating, place the emergency switch on. Remove the electrical connector from the engine low oil pressure switch and observe engine operation. If the engine continues to operate, remove the electrical connector from the engine overspeed switch and observe engine operation. The following shall constitute failure of this test:

(a) The engine stops when the low oil pressure electrical connector is removed.
(b) The engine does not stop when the engine overspeed switch electrical connector is removed.

4.6.3.3.4 Turbine exhaust temperature. Start the engine and load in accordance with 3.6.2.2(b). Direct the heated recirculated air into the turbine air intake with ducts (do not exceed 150° F) and open both the compressed air and vacuum valves fully. Note maximum engine exhaust temperature. Nonconformance to 3.7.9.3 shall constitute failure of this test.

4.6.3.3.5 Turbine air blockage. Connect a water gage manometer to the vacuum switch pressure sensing line, and with all panels in place operate the utility element. Slowly block the turbine air intake filter. Measure the vacuum sensing line pressure when the warning light glows and when the turbine stops (do not exceed 20 in. wg vacuum). Nonconformance to 3.7.9.5 shall constitute failure of this test.

4.6.3.3.6 Turbine conbustion air restriction. Insert a tee in the tube which connects the turbine air intake vacuum switches to the turbine air intake plenum and connect a water manometer to the tee. With the engine operating, gradually block the turbine air intake filter. Determine the pressure at which the turbine air inlet warning light glows and the pressure at which the turbine stops. NOTE: Do not exceed a vacuum of 21 inches water gage. At completion of the test, examine the turbine air inlet duct and plenum for determination. One or more of the following shall constitute failure of this test:

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(a) Nonconformance to 3.7.9.5
(b) Permanent deformation of the turbine air inlet duct or plenum.

4.6.3.4 Electrical protection.

4.6.3.4.1 Generator overload protection. Conduct this test in accordance with MIL-STD-705, Method 512.2. Nonconformance to 3.11.7.1 shall constitute failure of this test.

4.6.3.4.2 Short circuit. Conduct this test in accordance with MIL-STD-705, Method 512.1. Nonconformance to 3.11.7.1 shall constitute failure of this test.

4.6.3.4.3 Undervoltage and overvoltage. Conduct this test in accordance with MIL-STD-705, Method 512.3. Nonconformance to 3.11.7.3 or 3.11.7.4 shall constitute failure of this test.

4.6.3.4.4 Underfrequency. Conduct this test in accordance with MIL-STD-705, Method 506.1. Nonconformance to 3.11.7.2 shall constitute failure of this test.

4.6.3.5 Electrical system check.

4.6.3.5.1 Electrical receptacle voltage. Apply power to each utility element electrical output receptacle. Check the voltage of each receptacle socket. One or more of the following shall constitute failure of this test:

(a) 60 Hz or 400 Hz ac line-to-neutral - voltage less than 115 or greater than 125.
(b) 60 Hz line-to-line - voltage less than 230 or greater than 250.
(c) 400 Hz ac line-to-line - voltage less than 200 or greater than 220.
(d) Dc line-to-ground - voltage less than 24 or greater than 30.

Note: In order to obtain voltage on the dc leg of the 60 Hz power-receptacle, the 400 Hz system output switch (S9) must be in the "closed" position and the 400 Hz system output contactor (K-12) must be tripped.

4.6.3.5.2 Phase sequence. Conduct a phase sequence test on each 3-phase power receptacle of the utility element in accordance with

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MIL-STD-705, Method 507.1. Nonconformance to 3.11.12 shall constitute failure of this test.

4.6.3.5.3 Lights. Check panel lights and indicator lights as follows:

(a) Check all panel illuminating lamps for operation. Failure of any panel light to glow when actuated shall constitute failure of this test.
(b) With the master switch in "run" position, check the press-to-test indicator lights. Failure of the indicator lights to glow when checked shall constitute failure of this test.

4.6.3.6 Electrical system performance.

4.6.3.6.1 Voltage adjustment. Conduct the voltage adjustment test for both the 60 Hz and 400 Hz system in accordance with MIL-STD-705, Method 511.1. Rated load for this test shall be in accordance with 3.6.2.2(f). Nonconformance to 3.11.4 shall constitute failure of this test.

4.6.3.6.2 Voltage and frequency regulation. Conduct a voltage and frequency regulation test in accordance with MIL-STD-705, Method 614.1 for both the 400 Hz and 60 Hz generators. Rated load for this test shall be in accordance with 3.6.2.2 (f). Nonconformance to 3.11.1, 3.11.4, or 3.11.5 shall constitute failure of this test.

4.6.3.6.3 Transient voltage and frequency regulation. Conduct a transient voltage regulation test in accordance with MIL-STD-705, Method 608.1 for both the 400 Hz and 60 Hz generator systems. Rated load for this test shall be in accordance with 3.6.2.2(f). Nonconformance to 3.11.3 or 3.11.5 shall constitute failure of this test.

4.6.3.6.4 Voltage dip. Conduct a voltage dip test in accordance with MIL-STD-705, Method 619.1, except that the loads specified in 3.6.2.3 shall be used instead of the loads specified in MIL-STD-705, Method 619.1. Nonconformance to 3.11.3 shall constitute failure of this test.

4.6.3.6.5 Battery charger check. Measure the terminal voltage of the battery (or two batteries, if connected in series) when the ampere charging rate indicates zero (fully charged). Nonconformance to 3.11.9 shall constitute failure of this test.

4.6.3.7 Generators.

4.6.3.7.1 Wave form. Conduct a wave form test in accordance with MIL-STD-705, Method 601.1. The test shall be conducted at loads of zero, 50 percent and 100 percent of rated load at unity and 0.8 P.F.

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for both the 60 Hz and 400 Hz generator systems. Nonconformance to 3.11.2 shall constitute failure of this test.

4.6.3.7.2 Waveform harmonic. Perform a harmonic analysis of the voltage waveform of both the 60 and 400 Hz generator systems in accordance with MIL-STD-705, Method 601.4. Nonconformance to 3.11.2 shall constitute failure of this test.

4.6.3.8 Instrument check. All utility element instruments shall be tested in accordance with MIL-STD-705, Method 513.1. In addition, the engine oil pressure, fuel pressure and tachometer generator shall be checked by installing appropriate indicating instruments. This check may be conducted as part of another test. One or more of the following shall constitute failure of this test:

(a) Nonconformance to 3.11.8.
(b) A variation of more than plus or minus 2.5 percent in elapsed time when the hourmeter is compared to a stopwatch.
(c) A variation of plus or minus 5 percent when the exhaust temperature is compared to the master instrument.
(d) Inability to connect an oil pressure gage or fuel pressure gage to the utility element.
(e) Failure of the tachometer generator to give an engine speed signal.

4.6.3.9 Refrigeration system quality.

4.6.3.9.1 Refrigerant leak. Conduct this test with the utility element exposed to an ambient temperature which will produce a refrigerant pressure of 25 psig or greater with a halogen-sensitive electronic leak detector (General Electric Type H-2 or equal). Calibrate the leak detector for R-114 and set the detector to indicate a leakage rate of 0.05 ounce per year. Check completely both refrigerant systems for leaks. Any detectable leak shall constitute failure of this test.

4.6.3.9.2 Refrigeration moisture content. After not less than 1 hour of continuous operation on the cooling cycle in a minimum ambient temperature of plus 95° F, shut down the air conditioners. After an additional period of 1 hour, withdraw a sample of refrigerant vapor from the low pressure side and determine the moisture content by one of the methods specified in BB-F-1421. A moisture content of more than 15 ppm shall constitute failure of this test.

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4.6.3.10 Air conditioning system protection and controls.

4.6.3.10.1 Refrigerant high limit pressure switch. Attach a refrigerant pressure gage to the air conditioning System No. 1 high pressure side and operate the system with the thermostat set at maximum cooling. Gradually block off the condenser airflow until the compressor cuts off. Repeat tests for System No. 2. Nonconformance to 3.12.3.1.2 shall constitute failure of this test.

4.6.3.10.2 Cooling temperature control. Operate the utility element in an ambient temperature of 70° F to 85° F. Set the mode switch to "cool", the air conditioning select switch to No. 1 and the cooling thermostat to minimum cooling. Observe the conditioned air inlet and outlet temperatures. Rotate the thermostat setting to maximum cooling and observe the inlet and outlet temperatures. Repeat this check for air conditioning System No. 2. One or more of the following shall constitute failure of this test:

(a) An inlet air temperature/outlet air temperature difference of more than 5° F in minimum cool.
(b) An inlet air temperature/outlet air temperature difference of less than 10° F within 2 minutes after rotating the thermostat from minimum cooling to maximum cooling.
(c) Nonconformance to 3.12.2.2.

4.6.3.10.3 Evaporator anti-ice. Operate air conditioning System No. 1 in an ambient temperature above 70° F with the thermostat set in maximum cooling. Unfasten the anti-ice thermostat (P/N 13217E4063) from its mounting bracket (do not electrically disconnect) and immerse thermostat in a container of crushed ice. Failure of the air conditioning system to go into bypass (cool air indicating light not glowing) shall constitute failure of this test. Repeat for System No. 2.

4.6.3.10.4 Condenser door interlock. Attempt to start each air conditioning system as follows:

System Number	Condenser Cover Position	
	Intake	Discharge
1	Open	Closed
1	Closed	Open
2	Open	Closed
2	Closed	Open

Nonconformance to 3.12.3.1.1 shall constitute failure of this test.

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4.6.3.11 Air conditioning system.

4.6.3.11.1 Recirculating airflow check. Place test ducts on the recirculating air Outlets Nos. 1 and 3. The test ducts shall be 48 inches long and the same diameter as the recirculating air outlets. The end of the test duct away from the utility element shall be covered with a plate 1/8 inch thick containing a concentric hole 12 inches in diameter. Pressure taps shall be placed 90 degrees apart around the circumference of the duct and 1-1/2 inches from the plate end of the duct. Operate the utility element with covers on air Outlets Nos. 2 and 4, all covers off the four inlet openings, and fresh air dampers closed, in an ambient dry bulb temperature between 70° to 100° F and in vent modes 1 and 2. Measure the air pressure at the pressure taps for each duct with an inclined manometer conforming to the requirements of ASHRAE Standard 37. An air pressure for either air recirculating system 9 percent less than the air pressure determined by the approval preproduction Model No. 1 shall constitute failure of this test. The cooling system check shall not be conducted on a utility element until this test has been successfully completed.

4.6.3.11.2 Cooling system check. The cooling system check for System No. 1 and System No. 2 may be conducted simultaneously if the conditioned air outlet compartment divider panel is installed. Check the cooling capacity of System No. 1 by placing covers over Inlets 3 and 4 and Outlets 2, 3 and 4. Flexible ducts or other restrictions shall not be installed on either the inlet or outlet. Evenly space thermocouples in the location and of the quantity and type as follows:

Location	Minimum Quantity	Type
a. Inlet No. 1	8	Dry bulb
b. Inlet No. 1	4	Wet bulb
c. Inlet No. 2	8	Dry bulb
d. Inlet No. 2	4	Wet bulb
e. Outlet No. 2	8	Dry bulb
f. Condenser Inlet No. 1	18	Dry bulb

Set the mode select in cool, the air conditioning system select in position 1 and the System No. 1 cooling thermostat in maximum cool. Operate cooling System No. 1 until equilibrium is obtained, but for not less than 30 minutes before test data are recorded. Data shall then be recorded at 5 minute intervals until three consecutive sets of readings are within twice the tolerance prescribed in ASHRAE Standard 37.

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The inlet dry bulb temperature may be any selected temperature between 70° F to 100° F. The condenser air inlet temperature shall be within plus or minus 5° F of the cooling air inlet temperature (Note: A condenser air inlet temperature greater than plus 5° F is acceptable provided the air conditioning system meets the requirements of the this test). Test instrumentation shall conform to ASHRAE Standard 37 except the tolerance of temperature instrumentation shall not exceed twice that specified. For the period of equilibrium record the dry bulb and wet temperatures listed above, the air conditioning system power (kw) and the receiver liquid level (once only). The final temperatures shall be an average of those obtained during the equilibrium period. Inlets No. 1 and No. 2 temperatures shall also be averaged for a final temperature. Repeat the cooling capacity test of System No. 1 except place covers over Inlets 1 and 2 and Outlets 1, 2 and 4. Thermocouples or thermometers shall be installed as for System No. 1 except at Inlets 3 and 4; Outlet No. 3 and condenser air inlet for System No. 2. Record data as for System No. 1. One or more of the following shall constitute failure of this test:

(a) A temperature difference less than the temperature difference shown on Figure 1 for either System No. 1 or No. 2.
(b) Utility element power consumption exceeding the following limits:

Condenser Air Inlet Temp.	Maximum kw
70° F or less	25
80° F or less	26.3
90° F or less	27.5
100° F or less	29

(c) A liquid refrigerant level below the lower receiver sight glass.

4.6.3.11.3 Cooling capacity. Conduct a cooling capacity test in accordance with ASHRAE Standard 37 at the conditions specified in 3.6.1.1 with the following exceptions or stipulations:

(a) Control of the turbine air inlet temperature is not a requirement.
(b) The capacity test of each system may be conducted separately but both systems shall be in full cooling operation during the test and the external static pressure specified in 3.6.1.1 shall be imposed on both systems. The cooling capacity shall

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be the addition of the capacity obtained for each individual system.

(c) The air enthalphy method on the indoor side with a loop air enthalphy arrangement shall be used. The loop may be totally or partially located in the outdoor side test room and exposed to ambient temperature. The temperature of the air surrounding the loop shall be stable during the test and shall not vary at any point by more than plus or minus 5° F from the average. Corrections for the duct heat gain shall be calculated.

(d) The minimum size outdoor chamber shall be 30 feet by 24 feet by 14 feet high and the utility element shall be located in the chamber so that the minimum distance from any utility element side to the wall is not less than 8 feet. An unobstructive exhaust stack with a diameter of 16 inches and a length not exceeding 14 feet long may be used on the element. The hot water heater exhaust may be vented into the side of this stack.

(e) Recirculating air temperature measurements shall be made with a suitable sampling divice. The sampling device shall be installed inside the duct conveying the air being measured. Glass thermometers shall be used for the dry bulb and wet bulb temperature measurements and suitable window shall be used in the duct for reading the thermometers. The intake temperature samples shall be located inside the common duct section located before the intake plenums and the discharge air temperature samples shall be located in the duct before the airflow measuring device.

(f) The ambient air temperature shall be measured at the intake to the condenser. The measurement shall be made with a calibrated thermocouple grid constructed of 18 thermocouples evenly spaced over the condenser air opening.

(g) Record the following data in addition to that required by ASHRAE Standard 37:

(1) Condensing pressure, psig (at receiver).
(2) Evaporation pressure, psig (suction line).
(3) Refrigerant vapor temperature, °F (suction line near coil outlet).
(4) Refrigerant liquid temperature, °F (liquid line between hot gas defrost connection and evaporator coil.

Nonconformance to 3.6.1.1 or 3.6.1.2 shall constitute failure of this test.

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4.6.3.11.4 Evaporator coil frost. Operate the utility element for a period of 8 hours in full air conditioning (System Nos. 1 and 2) with the thermostat set so that the air conditioning systems do not cycle into bypass, in an ambient temperature of 80° F, plus or minus 10° F dry bulb, with air entering the air conditioner inlet at 65° F, plus or minus 3° F dry bulb, and a relative humidity not less than 80 percent. The recirculating air external static pressure shall be not less than three wg. One or more of the following shall constitute failure of this test:

(a) Damage occurring to the utility element during the test.
(b) Nonconformance to 3.12.3.

4.6.3.12 Heating system protection, control and performance.

4.6.3.12.1 Heating overtemperature switch. With the mode select switch in heat, and the No. 1 heat range switch in high and No. 2 heat range switch "off" block off the recirculating outlet, observe the maximum outlet temperature. Repeat for System No. 2. An outlet temperature less than 180° F or greater than 225° F shall constitute failure of this test.

4.6.3.12.2 Heating temperature control. Operate the utility element in an ambient temperature between 20° F to 70° F. Remove all covers from the recirculating system outlet and set the mode select switch in heat, operate for a minimum of 5 minutes with each of the following heat system select switch and heat range select switch settings:

Check	Heat Sys Select Sw	Heat Range Select Sw Sys 1	Heat Range Select Sw Sys 2	Thermostat Sys 1 and 2	Heat Control Valve Position Sys 1*	Heat Control Valve Position Sys 2
a.	1	OFF	OFF	MAX INCREASE	C	C
b.	1	LOW	OFF	MAX INCREASE	1/2-0	C
c.	1	MED	OFF	MAX INCREASE	3/4-0	C
d.	1	HIGH	OFF	MAX INCREASE	0	C
e.	2	OFF	LOW	MAX INCREASE	C	1/2-0
f.	2	OFF	MED	MAX INCREASE	C	3/4-0
g.	2	OFF	HIGH	MAX INCREASE	C	0
h.	1 & 2	HIGH	HIGH	MAX INCREASE	0	0
i.	1 & 2	HIGH	HIGH	MAX DECREASE	C	C

*C = Valve in closed position
0 = Valve in open position
1/2-0 = Valve one half open, etc.

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At the end of each 5 minute period operation check, measure the recirculating air inlet temperature and outlet temperature. One or more of the following shall constitute failure of this test:

(a) The heat control valves not in positions indicated for each check.
(b) A recirculating air temperature rise over 5° for any heating system when the heat control valve is in the closed position.
(c) A recirculating air temperature rise less than 30° F when the heat control valves of both systems are full open.

4.6.3.12.3 Electrical power failure during heat cycle. Operate the utility element with heat system 1 and 2 producing heat. Open the main contactor of the 400 Hz electric system to simulate power failure. Nonconformance to 3.12.4.1.2 shall constitute failure of this test.

4.6.3.12.4 Heating capacity. The heating capacity test shall be performed using the airflow measuring apparatus described in ASHRAE performed using the airflow measuring apparatus described in ASHRAE Standard 37 and the thermocouple grids for temperature measurement of the recirculating inlet air temperature and outlet air temperature. The external static pressure (recirculating air) shall be as specified in 3.6.1.3. All instrumentation and test apparatus used shall conform to ASHRAE Standard 37, except the tolerance of temperature may be twice that specified. One thermocouple per each 25 square inches of outlet and inlet area shall be used to measure the recirculating air temperatures; all thermocouples shall be located so that they are not affected by radiation. The test shall be started only after a state of equilibrium has been reached. Such a state of equilibrium shall be considered to exist if for a period of at least 30 minutes the inlet dry bulb temperature does not vary from the mean dry bulb temperature by more than 2° F and if the airflow rate does not vary by more than 2 percent. The test shall be conducted for not less than 1/2 hour. A minimum of four sets of readings shall be recorded. Both heating systems shall be tested simultaneously. The nonwinterized utility element shall be tested in an ambient temperature of minus 25° F and the winterized utility element shall be tested in an ambient temperature of minus 65° F. The recirculating air inlet temperature for both the nonwinterized and winterized utility element shall be selected from the range of plus 50° F to 65° F. The external static pressure (recirculating) shall be against 2 inches wg for all heating test. The airflow rate shall be calculated in accordance with ASHRAE Standard 37. The heating capacity shall be calculated from the airflow and the average dry bulb

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temperature difference between inlet and outlet recirculating air in accordance with ASHRAE Standard 37. Nonconformance to 3.6.1.3 shall constitute failure of this test.

4.6.3.13 Water system. Connect a supply of clean fresh water at atmospheric pressure to the utility element using the element supply hose. Connect straight tubes .45 plus or minus .05 inch inside diameter, 3 feet long and with a valve on the end to both the cold water outlet and the hot water outlet. One foot from the utility element, each tube should have a tap for a pressure gage. The tap opening into the tube shall be 1/16 inch diameter and the inside surface at the opening shall be smoooth and free of burrs. Start and operate the utility element. Prime the water system and place the cold and hot water pumps in operation. Operate the water system until temperatures are stabilized.

4.6.3.13.1 Cold water flow rate. Adjust the valve at the cold water tube outlet so that the outlet pressure is 10 psig plus or minus 0.5 psig. Measure the water flow by weighing the water discharge for a 5 minute period. Determine the flow rate in gallons per minute (gpm). Nonconformance to 3.6.3.1 shall constitute failure of this test.

4.6.3.13.2 Hot water flow rate. Determine the hot water flow rate following the procedure for the cold water flow rate. Nonconformance to 3.6.3.2 shall constitute failure of this test.

4.6.3.13.3 Hot water temperature control. Adjust the hot water flow rate at 0.9 to 1.0 gpm and measure the temperature of the hot water at the outlet. Nonconformance to 3.6.3.2 shall constitute failure of this test.

4.6.3.13.4 Hot water pump pressure switch. Check the hot water pump pressure switch by turning the cold water pump switch to "off" and open the cold and hot water outlet valves which were inserted in hoses. Failure of the hot water pump to shut off or the hot water heat supply valve to shut off when the cold water pressure drops to 10 plus or minus 2 pounds per square inch gage (psig) shall constitute failure of this test.

4.6.3.13.5 Hot water capacity. Connect the water system to an external source of water as specified in 4.6.3.13. Turn on both the hot and cold water pumps. Load the utility element so that the gas turbine exhaust temperature is at 950° F plus or minus 50° F. Lower the water temperature to 40° F plus or minus 5° F and adjust the hot

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water flow from the operating element to 1 plus 0.1 gallon per minute. Measure the minimum water temperature obtained for a period of 30 minutes. Nonconformance to 3.6.3.2 or 3.13.2 shall constitute failure of this test.

4.6.3.13.6 Water compartment heating. With the utility element operating, place ice on the water compartment temperature control. Determine if bleed air is emitted to the water compartment. The lack of bleed air blowing into the water compartment shall constitute failure of this test.

4.6.3.14 Compressed air. Utilizing the ejector assembly and turbine bleed air, measure the compressed air as capacity of the utility element at the rating conditions specified in 3.6.4. Use the instruments, apparatus and procedures specified in AMCA Bulletin No. 210. Nonconformance to 3.6.4 shall constitute failure of this test.

4.6.3.15 Vacuum system. The vacuum system capacity test shall be conducted using the instruments, apparatus and procedures specified in AMCA Bulletin No. 210. Nonconformance to 3.6.5 shall constitute failure of this test.

4.6.3.16 Winterization kit. Install the winterization kit and check for compatibility with a nonwinterized element. Check the operation of the electric heaters when the utility element is in an appropriate ambient temperature. One or more of the following shall constitute failure of this test:

(a) Winterization kit does not fit a nonwinterized element.
(b) Winterization electric heaters do not function.

4.6.3.17 Water leakage. With the utility element setting on a level surface, seal up the No. 1 and No. 2 condenser compartment floors and the refrigeration compartment floor drain and fill the floors with at least 3/4 inch of water. Observe the under surface of these compartment floors for water leakage. Nonconformance to 3.24.7 shall constitute failure of this test.

4.6.3.18 Environmental. Environmental tests shall include the following:

Sequence	Test Paragraph	Test Title
1	4.6.3.18.1	High Temperature
2	4.6.3.18.2	Low Temperature

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Sequence	Test Paragraph	Test Title
3	4.6.3.18.3	Rain
4	4.6.3.18.4	Humidity
5	4.6.3.18.5	Salt Fog
6	4.6.3.18.6	Roadability
7	4.6.3.18.7	Transit Drop
8	4.6.3.18.8	Rail Impact

Unless otherwise specified herein, the utility element for nonoperational tests shall have all doors and panels closed and fastened, the cover shall be fastened, the fire extinguisher and all accessories shall be stowed as shown on the decals of each utility element compartment. An environmental performance test shall be conducted before conducting any environmental test. The environmental performance test shall also be repeated after completing each environmental test. The environmental performance test shall consist of the following tests:

Test Paragraph	Test Title
4.6.3.2	Operation
4.6.3.5.1	Electrical receptacle voltage
4.6.3.6.2	Voltage and frequency regulation
4.6.3.9.1	Refrigerant leak
4.6.3.10.2	Cooling temperature control
4.6.3.11.2	Cooling system check
4.6.3.12.2	Heating temperature control
4.6.3.13.1	Cold water flow rate
4.6.3.13.2	Hot water flow rate
4.6.3.13.3	Hot water temperature control

4.6.3.18.1 High temperature. Conduct the test in accordance with MIL-STD-810, Method 501, Procedure I. The chamber temperature shall be stabilized and the utility element shall be operated (Steps 4 and 5) as follows:

(a) The chamber temperature (Step 4) shall be stabilized at 115° F. After stabilization, start the gas turbine and operate the gas turbine for 3 minutes. After operation stop the gas turbine.

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(b) The chamber temperature (Step 4) shall be stabilized at 140° F. Install flexible duct on all inlet and outlet recirculating air openings and provide means for measuring the recirculating air inlet dry bulb temperature at all openings. Start the utility element and both air conditioning systems; maintaining an ambient air temperature, turbine air inlet temperature and condenser air inlet dry bulb temperature of 140° F, plus 0° F, minus 4° F. The recirculating air inlet dry bulb temperature shall be decreased during air conditioner operation in accordance with the following schedule:

Elapsed operating time, min.	Recirculating air inlet dry bulb temp. °F.
15	120 ± 2.5
30	100 ± 2.5
45	90 ± 2.5
60 to end of test	75 to 85

After 1 hour elapsed time apply external electrical loads of 25 kilowatts (kw), 3-phase, 208 volt, 400 Hz power and 9.6 kw, 1-phase, 240 volt, 60 Hz power and maintain these loads until the end of the test. Continue operating the utility element for another 3 hours.

(c) The chamber temperature shall be stabilized at 120° F plus 5° F (Step 4). After stabilization, start and operate the utility element for a period of not less than 2 hours maintaining the ambient temperature at 120° F plus 5° F. The load specified in 3.6.2.2(a) shall be applied immediately after starting. One length of flexible duct shall be installed on each recirculating air inlet and outlet. During the operational period conduct the test specified in 4.6.3.6.2.

One or more of the following shall constitute failure of this test:

(a) Nonconformance to 3.5.1.
(b) Nonconformance to 3.6.2.2
(c) Failure of any environmental performance test.
(d) Any evidence of physical damage such as, but not limited to, loose gaskets, distorted bellows and diaphragms shall constitute failure of this test.

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4.6.3.18.2 Low temperature. Conduct a low temperature test in accordance with MIL-STD-810, Method 502, Procedure I. The storage temperature for the utility element shall be minus 65° F for a period of 36 hours (Step 2). The lowest operating temperature (Step 4) for a nonwinterized element shall be minus 25° F and for the winterized element shall be minus 65° F. The period of operation at the lowest operating temperature for either the nonwinterized or winterized utility element shall not be less than 8 hours. During the first hour of operation for either the nonwinterized or winterized utility element, perform the water system operation test (see 4.6.3.13) and the heating temperature control test (see 4.6.3.12.2) at the prevailing ambient temperature, with the exception of check (4) which shall be deleted. The hot and cold water pumps shall operate continuously during the test, but the water shall not be used externally except for specified tests. During the last 2 hours of operation, conduct the hot water temperature test (see 4.6.3.13.3), the voltage and frequency regulation test (see 4.6.3.6.2), the transient voltage regulation test (see 4.6.3.6.3), and the heating temperature control test (see 4.6.3.12.2). Stabilization of the generators for the voltage and frequency regulation and transient voltage regulation tests may be obtained by operation at loading (f) of 3.6.2.2 for 1 hour. When other tests are not being conducted, the utility element shall be operated at loading (b) of 3.6.2.2. The storage and operation sequences of the low temperature tests may be conducted in any order as required by the low temperature chamber schedule. All utility element panel instrument readings shall be recorded every half hour of operation. One or more of the following shall constitute failure of this test:

(a) Nonconformance to 3.5.3 or 3.7.3.8.
(b) Failure of any of the above tests.
(c) Failure of any environmental performance test.
(d) Failure of the utility element to maintain the specified electric load.
(e) Any physical damage.

4.6.3.18.3 Rain. This test shall consist of two phases. Phase I shall be conducted with the utility element not operating during exposure and Phase II shall be with the utility element operating continuously during exposure. The test item shall be the utility element complete with all accessories.

4.6.3.18.3.1 Phase I. With all accessories stowed in the utility element, conduct the rain test in accordance with MIL-STD-810, Method 506, Procedure I. After completion of the exposure period, examine the utility element for water penetration and water damage. Measure resistance between current carrying and current noncarrying parts with a

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megger. After completion of the examination and megger check, proceed with Phase II.

4.6.3.18.3.2 Phase II. Install one length of duct on all recirculating air openings and position the duct so that water will not enter the duct. Fully open the ventilation dampers. Start and operate the utility element at Loading Condition (a) of 3.6.2.2. With the utility element operating continuously, repeat the Phase I exposure.

One or more of the following shall constitute failure of this test:

(a) Nonconformance to 3.5.4.
(b) Failure of any environmental performance test.
(c) Water accumulation in the utility element.
(d) Evidence of water dropping on electrical controls.
(e) Evidence of water damage to insulation or internal parts.
(f) A resistance between current carrying and current noncarrying parts less than 50,000 ohms.

4.6.3.18.4 Humidity. Conduct the humidity test in accordance with MIL-STD-810, Method 507, Procedure I. One or more of the following shall constitute failure of this test:

(a) Nonconformance to 3.5.6.
(b) Failure of any environmental performance test.
(c) Corrosion or deterioration of any shaft, screw or any other hardware which prevents removal of equipment during servicing.

4.6.3.18.5 Salt fog. Conduct the salt fog test in accordance with MIL-STD 810, Method 509, Procedure I. One or more of the following shall constitute failure of this test:

(a) Nonconformance to 3.5.5.
(b) Corrosion or deterioration of any shaft, screw or any other hardware which prevents revival of equipment during servicing.
(c) Failure of any environmental performance tests.
(d) Loss of or change on the calibration of any control or instrument.

4.6.3.18.6 Roadability. Conduct the roadability test with the utility element transported on a Craig Model M840 dolly set over a block road. The cross-country terrain shall be similar to the Perryman No. 1 course at Aberdeen Proving Grounds, Maryland, consisting of native soil, dirt, mud and dust. The granite block road shall simulate a

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cobblestone road which provides an irregular and bumpy surface. The individual blocks shall average 5 inches in width. The blocks shall be laid on a concrete bed so the irregularities vary both along the length and width of the course and have crests of about 3 inches. The crests shall be such that a vehicle traveling over the road shall be subjected to both pitching and rolling motions. The granite block road shall be similar to the Belgian block course at Aberdeen Proving Grounds, Maryland. With the utility element mounted on the dolly set, tow the utility element over the courses for four 400-mile cycles for a total of 1600 miles. Each cycle shall consist of the following:

(a) Gravel road, 100 miles (20 to 30 mph).
(b) Cross country, 200 miles (3 to 20 mph)
(c) Granite block, 100 miles (15 to 20 mph).

Note: The maximum towing velocity shall be maintained consistent with safe driving practice at all times and shall be maintained within the ranges listed for each road type.

One or more of the following shall constitute failure of the roadability test:

(a) Any evidence of structural damage, visual misalinement of components, dislocation of component parts, or breakage of lines.
(b) Failure of any environmental performance test.

4.6.3.18.7 Transit drop. Conduct this test in accordance with MIL-STD-810, Method 516, Procedure II, except that the height of the drop shall be 10 inches and edgewise drops shall be substituted for cornerwise drops. Nonconformance to 3.10.3 shall constitute failure of this test.

4.6.3.18.8 Rail impact. Conduct this test in accordance with MIL-STD-810, Method 516, Procedure VI, except the utility element shall be impacted once at speeds of 4, 6 and 8 miles per hour plus or minus 5 percent in the forward direction and once at speeds at 4, 6 and 8 miles per hour plus or minus 5 percent in the reverse direction (a total of 6 impacts). Nonconformance to 3.10.2 shall constitute failure of this test.

4.6.3.19 Handling and positioning.

4.6.3.19.1 Inclined operation. Conduct this test in accordance with MIL-STD-705, Method 660.1. Operate utility element for 15 minutes in

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each of the four positions, side to side and front to rear, at 10 and 15 degrees inclination respectively. Nonconformance to 3.10.4 shall constitute failure of this test.

4.6.3.19.2 Nonoperational inclination. Incline the utility element to 45 degrees in each of the four positions (side to side and front to back). Determine whether there is any leakage or spillage. Nonconformance to 3.10.4 shall constitute failure of this test.

4.6.3.19.3 Lifting and towing attachments. Pick up the utility element using the utility element sling (Drawing 13217E4450) and the four attachments located at the four upper corners of the enclosure. Attach a spring scale between the sling and lift. Record the weight of the utility element. Secure the utility element on a fixed base and, using the same spring scale and lifting arrangement, apply a vertical force of 4-1/2 times the weight of the utility element. Any evidence of breakage or permanent deformation of the lifting attachments shall constitute failure of this test. After the completion of the lifting attachment test, subject the towing attachments to the forces and in the directions specified in 3.10.5. Any evidence of breakage or permanent deformation of the towing attachments shall constitute failure of this test.

4.6.3.20 Sound level measurement. Measure the sound level at eight locations in increments of 45 degree azimuth angles through 360 degrees at a length of 65 inches above ground level at a distance of 20 feet from the external contour of the utility element in accordance with MIL-STD-1474 for steady-state noise. Operate the utility element at Load Condition (a) of 3.6.2.2 with one length of duct attached to each inlet and outlet and with the open end of the duct directed away from the microphone. Where the steady-state noise level exceeds 85 dB(A), the distance from the external contour of the utility element at which the noise level is equal to or below 85 dB(A) shall be determined. Nonconformance to 3.9 shall constitute failure of this test.

4.6.3.21 Electromagnetic interference. The utility element shall be tested in accordance with Test Methods R.E.02 (0.15 MHz to 1000 MHz) and A-CE 2 (1.5 MHz to 65 Mhz) for equipment class tactical support (Non CE) as shown in Table A-1 of MIL-STD 461, Notice 1, Appendix. The contractor shall furnish the contracting officer a test plan and test report as required by MIL-STD 461. Disapproval of the test report shall constitute failure of this test.

4.6.3.22 Production performance. Operate the utility element for a period of 50 hours in accordance with Table III. Grade JP-4

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fuel shall be used for this test. The ambient temperature shall be greater than 70° F with at least 50 percent of the operational time above 90° F for all runs utilizing the air conditioning system(s). Runs may be performed in any sequence, but once a run is started it shall be completed without stopping. The utility element shall be instrumented so as to record all data needed to evaluate the utility element. Routine and scheduled maintenance shall be performed as required.

Table III. 50 Hour Operational Profile

Run No.	Hours Operation	Loading 3.6.2.2	Other Requirements
1	2	c	Internal Loading only
2	2	c	None
3	4	d	Operate Air Conditioning System No. 1
4	4	d	Operate Air Conditioning System No. 2
5	4 (see Note 2)	e	Operate Heating System No. 1
6	4 (See Note 2)	e	Operate Heating System No. 2
7	18	a	See Note 1
8	8	b	See Note 3
9	4	f	See Note 4

NOTES:

1. The temperature controls shall cycle the air conditioning system(s) (cooling on and cooling off). Each system shall make at least four complete cycles per hour and the cool on time shall be approximately 2/3 of the cycle.
2. The heat range select switch shall be set for 1 hour at high, 1/2 hour at medium, and 1/2 hour at low.
3. The temperature controls shall cycle each heating system on and off. Each heating system shall make at least four complete cycles per hour. The heat on time shall be approximately 2/3 of the cycle.
4. The external electric loads (both 60 and 400 Hz) shall be dropped four times during this run. The load shall be picked up within 2 minutes after dropping the load.

4.6.3.23 Recording data. Record the following data at the end of every run. On Runs No. 7 and 8, record the data for both on and off operation.

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(a) All control panel instruments.
(b) Ambient temperature; (dry bulb and wet bulb, °F),
(c) Recirculating air temperature in and out of the utility element (heating or cooling runs only) (dry bulb, °F).
(d) Fuel consumption (Runs 7, 8, and 9 only) (lb/hr).
(e) Oil consumption (24 hour period) (lb/hr).

4.6.3.23.1 Post operation. After completion of the production performance test, conduct the following tests and examinations:

(a) Refrigerant leak (4.6.3.9.1).
(b) Examination (4.6.2) - Examination Nos. 120, 121, 130, 131, 132, 133, and in addition, the exhaust stack shall be removed and the inside of the exhaust system shall be examined for cracks.
(c) Heating temperature control (4.6.3.12.2).
(d) Voltage and frequency regulation (4.6.3.6.2).
(e) Cooling system check (4.6.3.11.2).
(f) Water system test (4.6.3.13.1 to 4.6.3.13.4, inclusive).

4.6.3.23.2 Criteria for failure. One or more of the following shall constitute failure of this test:

(a) Inability of the utility element to start and operate under the required loading.
(b) Failure of any component or system.
(c) Failure of any test or examination specified in 4.6.3.23.1.
(d) Fuel or oil consumption exceeding the requirements of 3.6.6.

4.6.3.24 Reliability. Using the MTBF specified in 3.14.1, the utility element shall be tested in accordance with Table IV with "accept" and "reject" criteria in accordance with MIL-STD-781, Test Plan VII. Continue testing until either an "accept" or "reject" decision is reached. A failure is defined as any malfunction which cannot be corrected within 60 minutes by adjustment, repair or replacement action using controls and on-equipment tools or parts available to the using element, and is a malfunction which causes or may cause one or more of the following:

(a) Failure to commence operation, cessation of operation, or degradation of performance below specified levels.
(b) Damage to the utility element by continued operation.
(c) Safety hazards to personnel.

4.6.3.24.1 Data. At the initiation of the reliability test, during every other 100 hour cycle of operation and after completing the

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reliability test, the data pertaining to the following test will be recorded to the tolerances specified:

Test Paragraph	Test Title
4.6.3.2	Operation
4.6.3.5.1	Electrical receptacle voltage
4.6.3.6.2	Voltage and frequency regulation
4.6.3.11.1	Recirculating airflow check
4.6.3.11.2	Cooling system check
4.6.3.12.2	Heating temperature control
4.6.3.13.1	Cold water flow rate
4.6.3.13.3	Hot water temperature control

In addition, once a day the following data will be recorded:

(a) All control panel instruments.
(b) Refrigerant level from sight glasses in receivers.
(c) Total hours, engine and air conditioning systems.
(d) Fuel consumption.
(e) Oil consumption (amount added if required).
(f) Ambient temperature, dry bulb and wet bulb.

4.6.3.24.2 Failure analysis. An analysis of each failure shall be accomplished which shall include the following information as a minimum:

(a) Failure number.
(b) Date of failure.
(c) Serial number of utility element.
(d) Description of failed part; part name, serial number.
(e) Hours of operation on utility element and failed part; both total accumulated hours and reliability test hours.
(f) Description of failure incident, including previous history, symptoms of failure, and mode of operation.
(g) Analysis of cause of failure.
(h) Corrective action for test continuation.

4.6.3.24.3 Criteria for failure. Nonconformance to 3.14.1 shall constitute failure of this test.

Table IV. 100 Hour Operational Profile

Run No.	Hour Operation	Loading 3.6.2.2	Other Requirements
1	4	f	50% External load; Note 6

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Table IV. 100 Hour Operational Profile cont'd.

Run No.	Hour Operation	Loading 3.6.2.2	Other Requirements
2	4	f	100% External load; Note 6
3	4	c	50% External load
4	4	c	75% External load
5	4	c	100% External load
6	8	d	75% External load; operate a.c.system No. 1; Note 1
7	8	d	75% External load; operate a.c. System No. 2; Note 1
8	4	e	75% External load; operate heat System No. 1
9	4	e	75% External load; operate heat System No. 2
10	16	a	50% External load; Note 1, 2, 4, 7
11	20	a	75% External load; Note 1, 2, 3, 4
12	8	a	100% External load; Note 1, 2
13	4	b	50% External load; Note 4, 5, 7
14	4	b	75% External load; Note 4, 5
15	4	b	100% External load; Note 5

NOTES:

1. Ambient temperature shall be greater than 70° F.
2. Ambient temperature shall be 90° F or greater for 50 percent of the run time.
3. The relative humidity of air entering the recirculation air inlets shall be 80 percent or greater for 50 percent of the run time.
4. The temperature control(s) shall cycle the air conditioning system(s) (cooling on and off) or the heating system(s) (on and off) so that each system makes four complete cycles per hour. The on time shall be approximately 2/3 of the cycle time.
5. The heat range select switch shall be set in high, medium, and low for even time durations of the test run.
6. The external electric loads (both 60 and 400 Hz) shall be dropped four times during the run. The load shall be picked up within 2 minutes after dropping the load.
7. Utilizing the ejector assembly and turbine bleed air, completely open the bleed valve for 1 hour.

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8. Fuels utilized shall be MIL-T-5624 (Grade JP-4), VV-F-800 (Type FF-2) and MIL-G-3056 (Type I). DF-2 fuel shall be used for every 3rd 100-hour cycle, except for Run 6 which shall use MIL-G-3056 (Type I) fuel. MIL-T-5624 (JP-4) fuel shall be used for all other operations.

4.6.3.25 Maintainability. Throughout the entire test program all units shall be serviced and maintained for all scheduled and unscheduled maintenance in accordance with the procedures outlined in the maintenance literature. In addition to the above, one unit shall be subjected to a complete evaluation for all operator and organizational services and a sampling for direct- and general-support services. During the evaluation, each item described in the technical manuals shall be serviced, adjusted, inspected, replaced, or repaired to the extent allowed in the maintenance literature for operator and organizational maintenance levels. In addition, a sampling of the items of the functional group contained in the maintenance allocation chart of the technical manuals shall be serviced, adjusted, inspected, replaced, or repaired to the extent allowed in the maintenance literature for direct- and general-support services. Personnel used for this evaluation shall have the skills assigned to MOS 52C20, Power Pack Generation Equipment Repairman, and all shall have additional training on the utility element. Nonconformance to 3.14.2 shall constitute failure of this test.

4.6.3.26 Human factors. The utility element shall be evaluated to determine that the human factors criteria have been met. Nonconformance to 3.22 shall constitute failure of this test.

4.6.3.27 Safety evaluation. The utility shall be examined prior to testing and evaluated throughout all testing for compliance to safety requirements. Nonconformance to 3.8 shall constitute failure of this test.

4.7 Inspection comparison. The Government may select utility elements at any time during the contract production period and subject these utility elements to the tests marked "X" in Column 5 of Table II to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government on utility elements selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production utility elements. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests

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contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of any utility element to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the utility element and shall be cause for refusal by the Government to continue acceptance of production utility elements until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on utility elements produced and accepted under the contract. Any deficiencies found as a result of inspection comparison inspection will be considered prima facie evidence that all utility elements accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 Preproduction pack inspection.

4.8.1.1 Examination. The preproduction pack shall be examined for the defects specified in 4.8.2.3. Presence of one or more defects shall be cause for rejection.

4.8.2 Quality conformance inspection of pack.

4.8.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.2.3 Examination. Samples selected in accordance with 4.8.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

146.	Materials, methods and containers not as specified for Level A. Each incorrect material, method or container shall be considered one defect.
147.	Unprotected exterior metal surfaces not coated with preservative as specified for Level A.

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148. Preserved finished surfaces not covered or wrapped as specified for Level A.
149. Water system not completely drained as specified.
150. Engine not preserved in accordance with the referenced document and exception as specified.
151. Batteries not placed and secured in the battery compartment for Level A.
152. Fire extinguisher not packaged and placed in the storage compartment within the housing.
153. Maintenance tools not preserved and packaged in accordance with the referenced document for Level A.
154. Loose components not packaged as specified for Level A.
155. Loose components not placed in their applicable storage compartments.
156. Receptacles, connections and ports not capped or covered as specified.
157. Access panels and doors not fastened and latched as specified.
158. Doors not secured with strapping as specified for Level A.
159. Openings not covered as specified for Level A.
160. Power electrical panel and water control panel not covered as specified for Level A.
161. Marking missing, illegible, incorrect, or incomplete.
162. Cover not in place and securely fastened.

5. PACKAGING.

5.1 Preproduction pack. The contractor shall furnish a preproduction pack for examination of the utility element within the time frame specified (see 6.2) to prove prior to starting production packaging that the applied preservation, packaging and marking comply with the preparation for delivery requirements of this specification. Examination shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.5). The preproduction pack may be accomplished utilizing either a preproduction utility element or a production utility element. When a preproduction utility element is utilized, any preservation and packaging shall be removed by the contractor at no expense to the Government when requested by the Government to facilitate comparison between the preproduction utility element and production utility elements.

5.2 Preservation and packaging. Preservation and packaging shall be Level A or C, as specified (see 6.2).

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5.2.1 Level A.

5.2.1.1 Unprotected metal surfaces. Unprotected exterior metal surfaces or components of the utility element, not specifically preserved herein, requiring the application of the contact preservative in accordance with MIL-P-116, shall be coated with preservative as follows. Unfinished surfaces shall be coated with Type P-1 preservative. Finished surfaces shall be coated with Type P-7 preservative, and the coated surfaces wrapped or covered with barrier material conforming to MIL-B-121, Grade A, Type II, Class 2, secured in place with tape conforming to PPP-T-60, Type IV PPP-T-75, Type I. The preservatives shall conform to the applicable specification listed in, and shall be applied in accordance with, MIL-P-116.

5.2.1.2 Water system. The water system shall be drained by the following procedure:

(a) Detach supply and drain water hose assembly from external water supply, then detach other end from water inlet connection in water box panel assembly on right side of enclosure. Drain hose, then coil and place in protected area.
(b) Place hot and cold water shutoff valves in water box panel assembly in closed position.
(c) Detach dual water hose assembly and hose heater receptacle from connections on applicable element of the utility element, then detach other end from outlet connections in water box panel assembly on enclosure. Drain hose assembly then coil and place in protected area.
(d) Open water drains, hot and cold water shutoff valves, and water inlet connection in enclosure water box panel assembly. Open vent valves at top of hot water surge tanks. Allow all water to drain from water system.
(e) After all water has drained from system, close vent valves on hot water and surge tanks. Connect one end of bleed air hose assembly to compressed air outlet on lower left side of enclosure, connect other end of hose assembly to cold water outlet connection in water box panel assembly.
(f) Open compressed air valve approximately 20 to 30 degrees to deliver small flow of compressed air into water system to purge system of residual water.

CAUTION

Observe water system tank and plumbing for excessive heating due to purge air. If external temperature of water system components exceeds

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safe touch limit (160° F maximum), throttle compressed airflow, or shut off airflow completely until system cools down. Excessive temperatures will damage the pump or valve seals.

(g) When all water has been purged, note which water connection is discharging maximum amount of air, then close the connection.
(h) Note which connection now has maximum amount of discharge airflow; close that connection.
(i) Continue purging operation until only one remains open, then shut off compressed air delivery, remove bleed air hose assembly, and close remaining connection.

5.2.1.3 Engine. The gas turbine engine shall be preserved in accordance with MIL-E-5607, except that the engine compressor section shall not be sprayed with corrosion-preventive oil. The following engine preservation procedure is furnished as a guide in accomplishing the preservation:

(a) Remove the external fuel supply hose assembly drain completely and seal hose ends.
(b) Remove drain plug from utility element fuel tank and allow fuel to drain from tank. Replace plug.
(c) Detach fuel line which connects engine fuel solenoid valve to fuel atomizer assembly at fuel solenoid valve. Connect a line or hose to fuel solenoid valve and place overboard to drain.
(d) Connect a supply of oil conforming to MIL-L-6081, Grade 1010, to boost pump inlet.

NOTE

It may be necessary to supply oil under slight pressure until pump gears pick up oil.

(e) Place fuel boost pump switch on instrument panel assembly in on position, then move master switch on instrument panel assembly to start position. Allow engine to rotate by starter motor action for 15 to 20 seconds until oil is noted coming from drain line installed at fuel solenoid valve. Stop engine by moving master switch to stop position.

CAUTION

DO NOT EXCEED STARTER MOTOR DUTY CYCLE.

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(f) Disconnect supply of oil from boost pump inlet and allow all oil to drain from fuel pumps and fuel lines.
(g) Remove drain line from engine fuel solenoid valve and reconnect fuel lines.
(h) Remove fuel filter element from fuel filter assembly and from the engine fuel control unit. Inspect both filter elements. If filter elements are clean and serviceable, reinstall elements. If new elements are required, dip new elements in oil conforming to MIL-L-6081, then install elements.
(i) Remove drain plug from oil tank and drain oil from tank. Allow oil to drain until all system oil has drained. Reinstall plug in tank.
(j) Remove oil filter element from engine oil pump assembly. Inspect filter element. If element is clean and serviceable, reinstall element. If a new element is required, dip new element in oil specified for engine lubrication system, then install element.

5.2.1.4 Batteries. The charged and dry batteries shall be placed and secured in the battery compartment. The cables shall be disconnected from the batteries and secured to the battery compartment.

5.2.1.5 Fire extinguisher. The fire extinguisher shall be packaged in a close-fitting box conforming to PPP-B-636, W6c, and placed in the storage compartment within the housing.

5.2.1.6 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts.

5.2.1.7 Maintenance tools. Maintenance tools shall be preserved and packaged in accordance with PPP-P-40, Level A.

5.2.1.8 Instructional material. Instructional material shall be preserved in accordance with MIL-P-116, Method IC-1 or IC-3 and placed in the compartment provided.

5.2.1.9 Loose components. Loose components, such as water hoses and electrical cables, shall be coiled and the coils secured with not less than three equally spaced ties of cotton tape or twine. Air conditioning ducts shall be secured on the racks provided. Repair parts, maintenance tools, and any other loose components requiring the protection afforded by a fiberboard box to prevent damage or to aid in

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segregation or storage shall be packaged in boxes conforming to PPP-B-636, W5c or W6c, as applicable to the weight of the components. All loose components shall be placed in their applicable storage compartments within the housing.

5.2.1.10 Receptacles, connections and ports. All receptacles, connections and ports shall be capped or covered as provided for.

5.2.1.11 Access panels and doors. The access panels and doors shall be secured to prevent pilferage by the use of strapping conforming to QQ-S-781, Type I or IV, regular duty, Finish B, Size 3/4 by 0.015 inches. The strapping shall encircle the utility element either lengthwise or girthwise, as applicable. Edge protectors shall be utilized on each edge and as required to prevent damage.

5.2.1.12 Cover. The fabric cover (see Drawing D13220E1146) shall be in the storage and shipping position and shall be securely fastened.

5.3.1.13 Power electrical panel and water control panel. The exposed front of the power electrical panel and the water control panel shall be covered with barrier material conforming to PPP-B-1055, Class E-2, and secured in place with tape conforming to PPP-T-75, Type I.

5.2.2 Level C. The water system and engine of the utility element shall be preserved as specified in 5.2.1 for Level A. The fire extinguisher shall be packaged in a close-fitting box conforming to PPP-B-636, Type CF, Class Domestic, variety and grade optional, and placed in the storage compartment within the housing. Preservation and packaging of all other components shall assure protection against deterioration and damage from the contractor to the initial destination. Components shall be placed and secured in their applicable compartments. Receptacles, connections and ports shall be capped or covered as provided for. Access panels and doors shall be securely fastened and latched.

5.3 Packing (see 6.9). The utility element preserved and packaged as specified in 5.2 shall not require packing.

5.4 Marking (see 6.9). In addition to any special marking required by the contract or order, marking for shipment and storage shall be in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The utility element, self-contained, transportable, is intended for use in providing a contamination-free and controlled environment in which medical, surgical and ancillary activity procedures and other supporting functions can be performed.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When a winterization kit is required (see 3.1.1).
(c) When special tools and equipment are required (see 3.1.2).
(d) Time frame required for submission of preproduction model, and number of utility elements required (see 3.2).
(e) When the Government will conduct any or all of the preproduc- model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and and tests will be conducted by the Government and which examination and test shall be conducted by the contractor (see 3.2).
(f) When initial production inspection is required and number of utility elements to be furnished, when applicable (see 3.3).
(g) Color of finish coat required (see 3.21).
(h) When winterization kit is to be removed if other than as specified (see Table II, Note 2).
(i) Time frame required for submission of the preproduction pack (see 5.1).
(j) When strapping shall be Finish A (see 5.2.1.11).
(k) Level of preservation and packaging required (see 5.2).

6.3 Preproduction model. Any changes or deviations of production utility elements from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish utility elements conforming to this specification.

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6.4 Definitions.

6.4.1 End Item Final Inspection Requirement (EIFIR). An EIFIR is a check list of quality characteristics prepared by the Government which the contractor must follow during his final inspection of the completed end item to verify complete and functional conformance to contract requirements prior to submission for final acceptance.

6.5 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, package, and mark the utility element in accordance with this specification.

6.6 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, lubrication orders, wiring diagrams, and contractor's maintenance and operation manual to be furnished with each utility element.

6.7 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of utility elements.

6.8 Electromagnetic interference.

(a) This utility element has been previously procured and qualified to MIL-STD-461, Notice 1, and MIL-STD-462, Notice 1, as specified in Sections 3 and 4 of this specification. Subsequent notices or revisions should not be made applicable to a procurement contract.

(b) The contracting officer should include the following in procurement documents:

"(1) Notices or revisions other than those listed below are not applicable:

MIL-STD-461A, Notice 1.
MIL-STD-462, Notice 1.

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(2) Bidders are cautioned against failing to properly take into consideration the radio frequency interference suppression/electromagnetic interference characteristics requirements for the end items in preparing their bids.

The foregoing are performance requirements only. Specific drawings which may be incorporated therein are intended as an example of a method which was sucessful in the past.

Since each offeror has latitude in selecting electrical components within the requirements of the drawings and specifications, and the electromagnetic emission and suppression characteristics may differ from those previously used, the Government does not expressly or implicitly warrant that performance requirements will be met if such drawings are followed and reliance upon such drawings is at the contractor's own risk."

6.9 Packing. Levels of packing have not been included as the utility element container has been designed to act as the shipping container. For purposes of preservation, packaging/packing level marking, the marking shall be A/A or C/A, as applicable.

6.10 Government-furnished property. The contracting officer should arrange to furnish the property specified in 3.23.

Custodian:
Army - ME

User activity:
Navy - MC

Preparing activity:
Army - ME

Project 6115-0182

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X-2127

NOTE: FOR ALL COOLING AIR INLET CONDITIONS WHICH ARE ABOVE THE DRY COIL LINE, USE THE DRY COIL LINE AND COOLING AIR INLET DRY BULB TEMP. TO FIND THE TEMP DIFFERENTIAL.

FIGURE D - COOLING CAPACITY CHECK (SINGLE SYS.)

DRY COIL LINE

100°F

90°F

80°F

70°F

COOLING AIR INLET DRY BULB TEMP. °F

COOLING AIR INLET WET BULB TEMP. °F

TEMP DIFFERENCE (INLET DRY BULB MINUS OUTLET DRY BULB) °F

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-P-52632D(ME) Power Plant, Utility: MUST Hospital System

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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